



CUD Mental Health Week

Putting the site on the school server, with the school's own domain

For the school's IT person
Time needed: about 1 hour
plus waiting for DNS and the certificate

Students' phones
scan the QR code



Your domain, HTTPS
Caddy (or nginx / IIS) on 80 and 443



The app
one Node.js process, port 3000,
this machine only



data/app.db
one database file

1 What this is

A small Thai-language website: a mood check-in, a flower quiz for the booth, a booth TV display, and a password-protected admin page for the student council. It is a single Node.js program (Next.js) with a one-file database. **Nothing else is needed:** no cloud account, no separate database server, and it makes no calls to outside services. Fonts are served by the app itself. The load is light: a few hundred students over a couple of hours.

It stores **no personal data**: no names, student IDs, IP addresses or free text. Only anonymous answers.

2 What we need from you

A machine	Ubuntu 22.04+ or Debian 12 recommended (Windows: see 9). Always on during event days. 2 vCPU, 4 GB RAM, 16 GB disk (the app itself needs a fraction of that; the room is for building it, see step 3).
A hostname	For example <code>mentalhealth.yourschool.ac.th</code> , with a DNS record pointing at the machine. It must stay the same for the whole campaign: the QR codes are printed once.
HTTPS	A certificate for that name (step 5).
Network	Ports 80 and 443 reachable by the students' network. Only if students use mobile data, that means the internet too. Port 3000 must not be exposed.
Access	Read access to the code (a GitHub deploy key, or a zip from the student council) and internet during setup only.

3 Install

Node.js **22 or newer** is required. On Ubuntu/Debian:

```
curl -fsSL https://deb.nodesource.com/setup_22.x | sudo -E bash -
sudo apt-get install -y nodejs git
sudo useradd --system --create-home --shell /usr/sbin/nologin mentalhealth
```

Add swap before building. Building the site (next) briefly needs well over a gigabyte of memory, more than running it day-to-day. A swap file means a memory spike slows the build down instead of killing it outright. Skip this only if the machine has 8 GB+ of its own memory:

```
sudo fallocate -l 2G /swapfile && sudo chmod 600 /swapfile
sudo mkswap /swapfile && sudo swapon /swapfile
echo '/swapfile none swap sw 0 0' | sudo tee -a /etc/fstab
```

Get the code into `/home/mentalhealth/app` (either way works):

```
sudo -u mentalhealth git clone
git@github.com:Artycud/MentalHealth.git
/home/mentalhealth/app
# or: unzip the project there, then: sudo chown -R
mentalhealth: /home/mentalhealth/app
```

```
cd /home/mentalhealth/app
sudo -u mentalhealth npm ci
sudo -u mentalhealth npm run build
```

4 Keep it running

There is nothing to configure. The council sets its own admin username and password in the browser, the first time they open `/admin` (step 6). The database defaults to `data/app.db` inside the project folder, so no settings file is needed; just make sure the folder is owned by the service user (the `useradd` above did it).

Create `/etc/systemd/system/mentalhealth.service`. This starts it on boot and restarts it if it stops:

```
[Unit]
Description=CUD Mental Health Week
After=network.target

[Service]
User=mentalhealth
WorkingDirectory=/home/mentalhealth/app
Environment=NODE_ENV=production
ExecStart=/usr/bin/node node_modules/next/dist/bin/next
start -H 127.0.0.1 -p 3000
Restart=always
RestartSec=3

[Install]
WantedBy=multi-user.target

sudo systemctl daemon-reload && sudo systemctl enable --now mentalhealth
```

5 Domain and HTTPS (Caddy)

Caddy is the simplest: it gets and renews the certificate itself. `sudo apt-get install -y caddy`, then `/etc/caddy/Caddyfile`:

```
mentalhealth.yourschool.ac.th {
    reverse_proxy 127.0.0.1:3000
}

sudo systemctl reload caddy
```

- **Automatic certificates** need public DNS for the name and ports 80/443 open from the internet.
- **Inside the school network only?** Use the school's own certificate instead: add `tls /path/cert.pem /path/key.pem` inside the block.
- nginx or IIS work equally well: proxy to `127.0.0.1:3000` and pass `X-Forwarded-Proto`. No WebSockets are used.
- The app sends its own security headers. After HTTPS is confirmed working, you may add HSTS: `header Strict-Transport-Security "max-age=31536000"`.

Do not put a login, captive portal or single sign-on in front of it. Students arrive from a QR code and must open the page directly.

6 Check it works

```
# expect HTTP 200:
curl -I https://mentalhealth.yourschool.ac.th/
sudo systemctl status mentalhealth
sudo journalctl -u mentalhealth -n 50 --no-pager
```

Address	What it is
/	Students land here (the QR code)
/booth/kiosk	The iPad at the booth
/booth/display	The TV (open full-screen)
/admin	Student council: statistics, dates, export
/booth/login	Sign-in for the iPad and TV, once the council has made a booth password in /admin

Open `/` on a phone over **mobile data** and over **school wifi**, if both are meant to work.

First time in the admin. When the council opens `/admin` it asks them to choose a username and password, and for a one-time **setup code** that only you can read (it stops a stranger claiming the account before them). Find it, and give it to the person setting up:

```
sudo journalctl -u mentalhealth | grep "Setup code"
# or: cat /home/mentalhealth/app/data/admin-setup-code-app.txt
```

7 Backups (please set this up)

All the data is one file. Copying it while the site is running can give a broken copy, so use the built-in command. It is safe at any moment and keeps the newest 60:

```
cd /home/mentalhealth/app && sudo -u mentalhealth npm run db:backup
```

Hourly, in `/etc/cron.d/mentalhealth`:

```
0 * * * * mentalhealth cd /home/mentalhealth/app && npm run db:backup
```

Snapshots appear in `data/backups/`. Also copy that folder to **another machine** once a day. **Restore:** `systemctl stop mentalhealth`, copy the chosen snapshot over `data/app.db` (delete any `app.db-wal` / `-shm` files beside it), then `systemctl start mentalhealth`.

8 Updating

Not during booth hours. It takes under a minute:

```
cd /home/mentalhealth/app && sudo -u mentalhealth git pull
sudo -u mentalhealth npm ci && sudo -u mentalhealth npm run build
sudo systemctl restart mentalhealth
```

The database is never touched by an update. New tables are added automatically on first start.

Project: github.com/Artycud/mentalHealth (private). Questions: ask the student council for the project owner.

9 Windows Server instead

Install Node.js 22 (MSI), then in the project folder run `npm ci` and `npm run build`. Run it as a service with [NSSM](#): program `node.exe`, arguments `node_modules\next\dist\bin\next start -H 127.0.0.1 -p 3000`, start directory the project folder, startup type Automatic. Put HTTPS in front with IIS (URL Rewrite + ARR) or Caddy for Windows. Schedule `npm run db:backup` in Task Scheduler every hour.

10 If something is wrong

Symptom	Look at
502 / blank page	The app is not running. <code>journalctl -u mentalhealth -n 50</code> . Usually Node older than 22 (<code>node -v</code>) or <code>npm run build</code> not run.
Cannot open the database	The <code>data/</code> folder must be owned by <code>mentalhealth</code> and writable.
Certificate error	DNS for the name is wrong, or ports 80/443 are blocked. See <code>journalctl -u caddy</code> .
Forgot the admin password	Signed-in admins can change it in the panel. If it is lost: <code>cd /home/mentalhealth/app && sudo -u mentalhealth npm run admin:reset</code> , type RESET, then open <code>/admin</code> and set a new one (a new setup code is in the log).
Locked out of admin	Five wrong passwords lock the login for 10 minutes. Wait, then retry.
Times look wrong	They should not: the app uses Thailand time itself, whatever the server's clock zone.

11 Good to know

- Keep the hostname unchanged for the whole campaign. QR codes are printed once and cannot be edited.
- The booth TV and kiosk are ordinary browser pages: no software to install on them.
- The council can switch the live booth, edit dates, export or delete data from `/admin`. You do not have to.
- When the report is done, the council can wipe all answers from `/admin`. Take a last backup first if they want to keep it.

Source of this page: [docs/school-server-guide.html](https://docs.school-server-guide.html)